



Spot the Wrong Block

Quick facilitation notes & answer key

Goal

Students locate a bug by comparing a plan (what each step should be) to the code (what is there) and finding the block with the wrong number. Reading existing code against its intended behaviour to find where it goes wrong is the core of C3.2 debugging.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 to run the code and the plan and compare where Bit lands.

Run it (about 20 min)

1. I DO: run Code A — 'start 1, forward 2 to 3, forward 3 would be 6 but the path ends at 5, so Bit stops on 5. The plan said land on 4 — so there is a bug.'
2. WE DO: Exercise 1 — Code A lands on 5. Exercise 2 — compare to the plan; the last block says forward 3 but the plan says forward 1, so circle forward 3.
3. YOU DO: Exercise 3 — Code B lands on 0; the last block says back 3 but the plan says back 1.
4. Connect: 'Find the bug by checking each block against the plan — the one that does not match is wrong.'

Answer key (modelled by the run-gate)

Ex 1 — Code A: 1 → forward 2 → 3 → forward 3 → stops at 5. Bit lands on 5, not the target 4.

Ex 2 — the wrong block is the last one, forward 3; it should be forward 1 (then 1 → 3 → 4).

Ex 3 — Code B: 5 → back 2 → 3 → back 3 → stops at 0. The wrong block is back 3; it should be back 1 (then 5 → 3 → 2).

Watch for & differentiate

Common slip: circling a block that matches the plan — have students check each block against the plan, one at a time.

Simplify: read the plan and the code aloud together, pausing at the block that doesn't match.

Extend: ask students to write the corrected Code A and run it to confirm it reaches 4.

Success indicator: the child finds Code A lands on 5 and Code B on 0, and circles forward 3 / back 3 as the wrong blocks with the right fix.